

Identifying encephalopathies from acute metabolic derangements

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Metabolic derangements, when acute and severe, affect brain function. This presents mostly with a marked decline in the level of consciousness, resulting in impaired responsiveness, abnormal reactivity, impaired content, and loss of memory retention. The term metabolic encephalopathy has been used but is conjecture that can be challenged in the age of modern neuroimaging. We now recognize that many metabolic encephalopathies may involve structural lesions and at an early stage. Common clinical conundrums are the evaluation of the degree of brain injury and its recoverabil-

ity. This review discusses the appropriate terminology for these conditions, the diagnostic approach, therapy recommendations, and prediction of recovery potential. In evaluating a presumed metabolic cause for encephalopathy, we must (1) search for and rule out structural injury to the brain despite an obvious explanatory metabolic derangement, (2) recognize that several confounding conditions often co-exist, and (3) acknowledge that resolution of brain dysfunction may be protracted despite normalization of laboratory values.

Keywords: electrolyte abnormalities, encephalopathy, magnetic resonance imaging (MRI), organ failure, outcome

Introduction

Any condition disrupting metabolic processes may indirectly affect brain function. Brain dysfunction is mostly seen in critical illness following major organ failure or a cascade of multiple organ failures. When the brain acutely fails, it affects the other organs. The brain and major organs have a reciprocal relationship. They seldom fail in isolation from each other. This sympathetic overdrive “stresses” the heart, lungs, and gastrointestinal organs. Conversely, any acute, clinically significant organ failure affects brain physiology. This brain dysfunction has been called metabolic because it uproots the fine-tuned, metabolic regulatory systems in the neurons. Recently, and far more often than previously appreciated, with apparent metabolic derangements, encephalopathy correlates to a structural brain injury. For example, magnetic resonance imaging (MRI) has shown widely distributed white-matter edema in severe uremia and diffuse cortical swelling in acute, fulminant hepatic failure. Once organ failure resolves, the previous level of brain function may not resume; specific structural injuries may cause ongoing deficits.

Although the list of neurologic conditions labeled “metabolic” seems endless, only a few critical clinical conditions in actual clinical practice cause the most pervasive organ dysfunction-related biochemical changes. How do clinicians categorize these conditions? This review examines the appropriate terminology, the diagnostic approach, therapy recommendations, and prediction of potential recovery. Common clinical conundrums include evaluating the degree of injury and recoverability. This review offers three fundamental precepts. First, the search for structural brain injury should continue despite an obvious explanatory metabolic derangement. Second, confounding conditions must be recognized and eliminated. Finally, resolution of brain dysfunction may be protracted despite normalized laboratory values.

The brain disease called metabolic

The commonly used clinical “diagnosis” of metabolic encephalopathy needs refinement. Encephalopathy has been a “catch-all” for conditions with perceptual abnormalities leading to inattention, distraction, and decreased consciousness.

Metabolic is applied to substrate abnormalities or cerebral enzymatic activity leading to neuronal dysfunction. Conditions are loosely defined as “metabolic” when they result from insufficient substrate and/or oxygen, interference with cerebral enzymatic activity, transport of metabolites into a nerve cell, and oxidative phosphorylation by unstimulated neuronal respiratory processes. Encephalopathy often correlates with organ failure, using qualifiers such as uremic encephalopathy, hepatic encephalopathy, pulmonary encephalopathy, pancreatic encephalopathy, and post-anoxic (or post cardiopulmonary resuscitation [CPR]) encephalopathy. Encephalopathy refers to a broad brain dysfunction and, when significant agitation occurs, is renamed delirium. Delirium qualifies as an encephalopathy except with additional extreme restlessness or exaltation with acoustic or visual hallucinations [1]. Because these neurologic states overlap or converge, some have proposed “acute brain failure,” a term that parallels widely accepted terms such as “acute kidney failure,” “acute respiratory failure,” and “acute cardiac failure.” Others further divided delirium into a state with drowsiness alone versus a state with marked decline in the level of consciousness [2].

Once we accept encephalopathy as a workable term, is it justified to pair it with the adjectives metabolic or toxic metabolic? We recognize the toxicity of metabolic derangements. Indeed, externally administered toxins may produce a similar encephalopathy with any known metabolic derangement. The risk is that additional qualifiers will dilute specificity and discourage the search for the major inciting process. Loosely diagnosing patients after a neurologic consultation with “toxic metabolic encephalopathy” (and, worse, adding the additional qualifier “multifactorial”) provides little help. Innumerable neurotoxic substances cause encephalopathies (e.g., environmental, organic, and industrial neurotoxins; select heavy metals; organic or industrial toxins, such as pesticides; snorted, smoked, or injected illicit drugs; and incidental or intentional poisoning). Diagnostic tests are unavailable for many neurotoxins. However, single exposures can cause permanent deficits or death. This is adding the qualifier “toxic” to encephalopathy that is best reserved for these agents, which comprise a separate neurotoxicology field not further considered here. In-hospital, drug-related neurotoxicity is unusual but will be mentioned when relevant.

Organ failure indirectly leads to neuronal dysfunction. Often, insufficient substrates (e.g., oxygen and glucose) or circulation of toxic intermediates are responsible, particularly in acute renal or lymphatic failure. Multi-organ failure often results from sepsis, caused by an inflammatory surge. The combination of several organs in failure does not, in itself, account for neuronal dysfunction. Similarly, the brain is often an “innocent bystander,” whereas an acute insult to neuronal function may cause permanent abnormalities and structural decline. Figure 1 summarizes the major pathophysiological mechanisms. Table 1 offers a general overview of common causes.

What confounds and confuses

Clinicians must recognize the many confounders in critical illness resulting from interventions (or no intervention). These include drug infusions, shock, severe electrolyte derangements, and antibiotic- and chemotherapy-associated toxicity.

Drug infusions

Patients with organ failure invariably receive polypharmacy. Several administered drugs are sedatives, and the delayed responsiveness confounds the neurologist’s assessment. Drug elimination in patients with acute renal failure can be protracted [3, 4]. Other factors—such as a laboratory outlier—confound the neurologic findings of encephalopathy. Older age, with decreased total body water, body mass, and serum albumin, leads to decreased volume of distribution, increased drug effect, and decreased drug elimination. Fentanyl, the most problematic drug, is ubiquitous in intensive care units (ICUs), sometimes with dubious indications. Opioid-related effects, with differences based on their intrinsic properties, are complex. Although onset is immediate and lasts for several hours, half-lives of fentanyl increase after a few days of infusion, and its clearance in critical illness may be 10 times slower than in patients without organ dysfunction or hemodynamic instability [4]. Neurologists who must incorporate its potential confounding effect consider it an “unfriendly” drug. Reversal agents help, but benzodiazepines or opioids do not dissipate easily.

Shock

Examination of a hemodynamically unstable patient on multiple vasopressors yields ambiguous results. Hypotension alters consciousness;

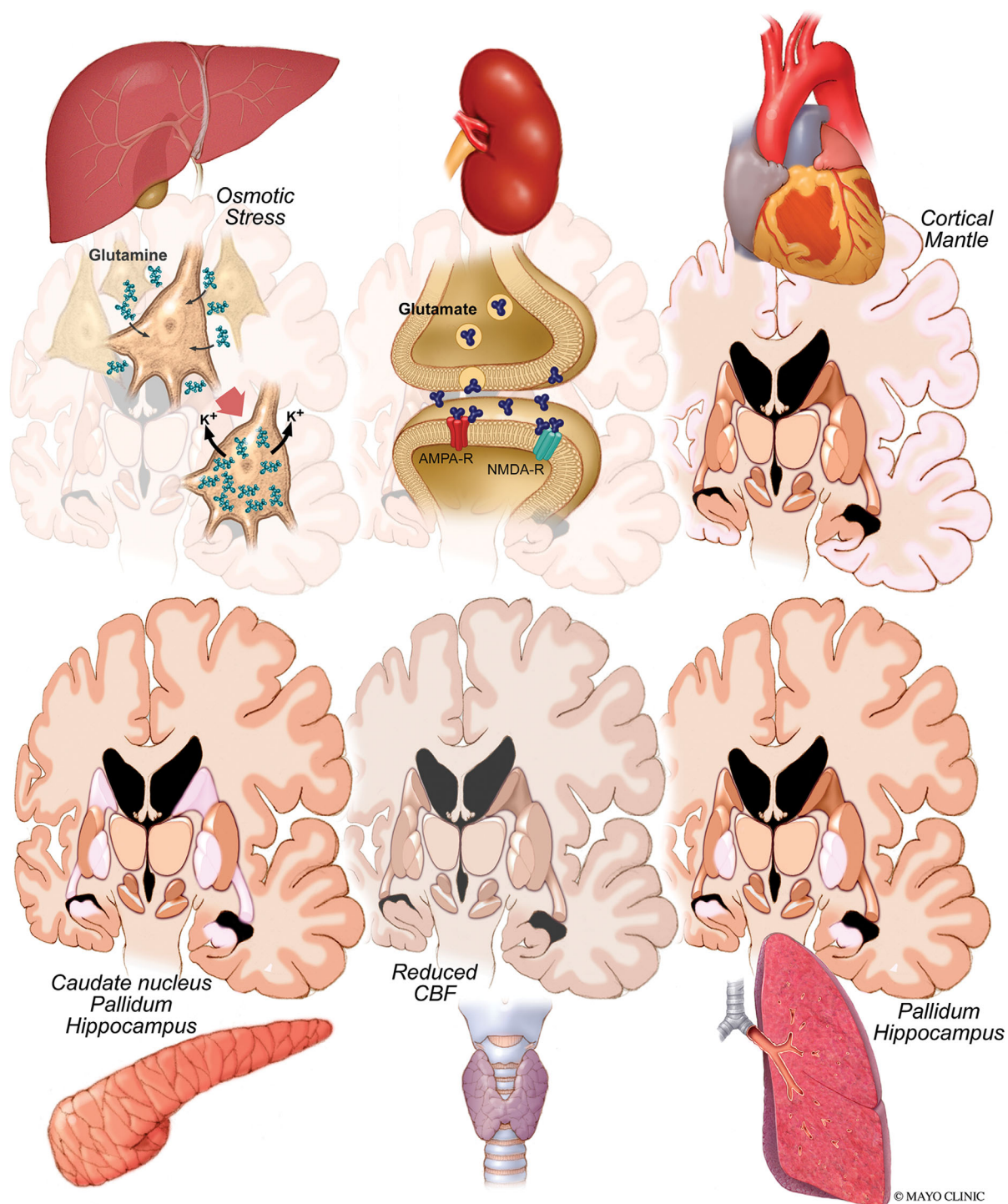


Fig. 1 Illustration of pathophysiological changes involving cerebral blood flow or enzymatic dysfunction with each organ failure.

Table 1. Common causes of acute encephalopathy^a

Acute shock
Acute azotemia
Acute hyperammonemia
Acute hyperglycemia
Acute hypoglycemia
Acute hyperthyroidism
Acute hypothyroidism
Acute vitamin B deficiency
Acute hypertension
Acute antibiotic toxicity
Acute chemotherapy toxicity
Any self-inflicted drug overdose

^aSee text for interventions.

the mean arterial pressure (MAP) with decreased consciousness is ~50 mmHg. Vulnerable areas for ischemia are in marginally perfused watershed territories; for the brain, a MAP threshold of <50 mmHg is considered ischemic (MAP thresholds are higher in other organ systems, for example, MAP for sufficient coronary perfusion pressure is 65 mmHg and 70–75 mmHg for a well-perfused kidney).

Electrolyte derangements

Most electrolyte abnormalities are immaterial to neurologists. Acute, severe metabolic derangements eventually reduce alertness to stupor. Localizing signs, for example, aphasia or hemiparesis, do not present with diffuse, metabolic derangement-induced brain dysfunction. One exception is nonketotic, hyperosmolar hyperglycemia occasionally causing transient aphasia or hemiparesis, although the mechanism is poorly understood.

Hyponatremia usually is asymptomatic until levels drop below 125 mmol/L. Although rapid sodium change is crucial, the degree of hyponatremia can also be an important factor. A precipitous drop to very low levels (<115 mmol/L) may cause cerebral edema and brain-tissue shift. Normally, the brain extrudes inorganic solubles within hours of hyponatremia followed by water loss, which reduces the risk of cellular swelling. However, rapid-onset edema overwhelms this compensatory system. Correcting severe hyponatremia (<120 mmol/L), particularly when it within 24 h, may cause demyelination of the pontine tracts (central pontine myelinolysis). Patients with long-standing alcohol-use disorder are most susceptible

to changes due to correction of chronic hyponatremia. Extrapontine myelinolysis affects the cerebellum, putamen, caudate nuclei, lateral geniculate bodies, hippocampus, cerebral cortex, and thalamus; it causes flaccid tetraparesis, bulbar palsy, dysarthria, dysphagia, and ocular motor abnormalities. In many patients, rapid correctly, chronic, severe hyponatremia is responsible, with “rapid” defined as (1) a 6 mmol/L correction of sodium within 6 h or (2) a 4–6 mmol/L increase in serum sodium concentration [5–7]. However, hyponatremia is different. A hyponatremia level ≥ 160 mmol/L confounds the clinical examination [8]. However, plasma osmolality may correlate better with consciousness levels than with an absolute hyponatremia value. Plasma osmolality, >350 mOsm/L, causes progressive drowsiness in other situations.

Hypocalcemia may induce neuropsychiatric symptoms such as subtle personality changes, concentration difficulties, confusion, depression, dementia, anxiety, and proximal muscle weakness. Levels <7.5 mg/dl cause several neuromuscular manifestations including circumoral numbness, tingling fingertips, muscle cramps, painful muscle spasms, or the more dangerous laryngospasm or bronchospasm. Chvostek and Trousseau signs are classic findings in hypocalcemia and remain useful in practice. In the Chvostek sign, tapping the skin over the facial nerve just anterior to the external auditory meatus causes facial-muscle contraction. In the Trousseau sign, inflating the blood-pressure cuff to 20 mm Hg above the patient’s systolic blood pressure causes carpal spasm after 3–5 min.

Antibiotic toxicity

Several frequently administered antibiotics unexpectedly reach toxic levels, producing major symptoms. The antibiotic metronidazole treats several bacterial and protozoal infections and may cause encephalopathy. It usually appears with prolonged (7–30 days) treatment [9, 10]. Patients have additional cerebellar symptoms, confusion, and seizures.

A recently discovered confounder is cefepime neurotoxicity. Cefepime crosses the blood–brain barrier, while sepsis and previous brain injury further disrupt the integrity of the barrier, causing increased central nervous system penetration ($\leq 45\%$; normal = 10%). Because cefepime is excreted renally (85% unchanged), reduced renal function proportionally increases the elimination

half-life and reduces the total-body clearance rate. Periodic sharp waves and triphasic waves characterize cefepime neurotoxicity. A recent study showed on electroencephalogram (EEG) periodic epileptiform discharges five times more frequently in patients receiving cefepime rather than meropenem. Overall, the prevalence of this finding was still relatively low (1.3%) [11].

A systemic review of cefepime toxicity found varying degrees of reduced consciousness, myoclonus, and confusion as the most common symptoms (approximately half). All EEG-tested patients had abnormalities, including nonconvulsive status epilepticus (1/4), but typically nonspecific, inconsequential triphasic waves and focal sharp waves [12]. Half were technically overdosed; however, one in four experienced neurotoxicity despite appropriate dosing. During continuous renal replacement treatment, limiting a cefepime dose to ≤ 2 g may decrease the risk of cefepime-induced neurotoxicity while reducing efficacy. Detected neurotoxicity requires switching to other antibiotics [11–14]. Myoclonus and asterixis do not necessarily derive from acute azotemia; this clinical setting is suggestive of cefepime neurotoxicity. Best practice is to hold cefepime and substitute another suitable antibiotic. Marked encephalographic improvement occurs within days.

Neurotoxicity of chemotherapy

Many chemotherapeutic agents correlate with acute, transient, toxic encephalopathy, as noted in a recent series [15]. The typical culprits are alkylating agents (e.g., cisplatin, thiotepa), antimetabolites (e.g., methotrexate, 5-fluorouracil), vinca alkaloids (e.g., vincristine), topoisomerase inhibitors (e.g., etoposide), DNA-intercalating agents (e.g., doxorubicin), immune checkpoint inhibitors, and C19-CAR-T-cell-immunotherapy-associated neurotoxicity. Diagnosis remains difficult because many patients present with an indistinct MRI. Cerebrospinal fluid (CSF) examination will show increased protein and predominant lymphocytic pleocytosis.

Typically, encephalopathic patients have decreased levels of consciousness, often with fever, headaches, limbic structural involvement, and a presentation resembling paraneoplastic encephalitis [16–18]. Empirical management involves high-dose corticosteroid infusions followed by plasma exchange, IVIG, and rituximab. Neurotoxicity of CD19-CAR-T-cell immunotherapy

ranges from mild to severe. Severe forms lead to rapid-onset encephalopathy, often with significant speech apraxia (distorted vowels, stumbling on complex words) or global aphasia (mute, no repetitions or comprehension). MRIs show white-matter edema. Paraphrasing errors, febrile perseveration, global aphasia, or mutism define severe neurotoxicity. Most neurotoxicity cases resolve within 3–8 weeks, although one in 10 patients needs long-term follow-up before showing further improvement. MRI often shows T2-FLAIR hyperintensities in bilateral thalami, brainstem, and cerebral white matter.

MRI can be helpful but nonspecific. Toxic and metabolic brain disorders display general imaging patterns [15]. These are (1) symmetric basal ganglia or thalami involvement in Wernicke encephalopathy, uremic encephalopathy, and severe hyperglycemia; (2) symmetric dentate nuclei involvement with metronidazole; (3) cortical gray-matter involvement with extreme hypoglycemia and hyperammonemia; (4) symmetric periventricular white-matter involvement (with gray-matter sparing) with chemotherapeutic agents; (5) corticospinal-tract involvement with cobalamin deficiency or vitamin B12 deficiency; (6) corpus callosum involvement with antiepileptic drugs; (7) asymmetric white-matter involvement (demyelinating disease pattern) with tumor necrosis factor- α blockers; (8) parieto-occipital subcortical vasogenic edema with posterior reversible encephalopathy syndrome (PRES); and (9) central pons involvement with an osmotic demyelination syndrome [15].

When organs fail

Specialists diagnose organ dysfunction. These clinicians may consult the neurologist (or neurointensivist) for a detailed assessment of the noted encephalopathy but also to exclude treatable, mimicking neurologic disorders. This section reviews the separate organ systems.

Acute pulmonary disease

The COVID pandemic has shown how acute pulmonary disease associated with hypoxic-hypocarbic failure leads to markedly changed consciousness levels. The SARS-COVID-2 acute respiratory syndrome leads to profound hypoxemia and increased respiratory drive, manifesting as increased tidal volume but not necessarily dyspnea. Patients with increased respiratory drive

require substantial sedation and, eventually, mechanical ventilation. The relationship between increased respiratory drive and worsening lung injury remains unknown. Angiotensin-mediated sensitivity of chronic bodies, which express ACE-2 receptors, may interact in the respiratory centers of the brainstem [19, 20].

The role of hypercapnia remains unclear but with numerous reported findings of cortical depression or dampened neuronal activity on EEG. Metabolic effects of hypercapnia include cerebrovascular dilatation (opposed to hypoxemia, which changes cerebral blood flow less frequently). When hypoxia occurs, the breathing rate increases and deepens. Changes in PO₂ must be profound because the carotid-artery baroreceptors are set at very low arterial PO₂ levels. Most patients are unaware of dyspnea with only moderate hypoxemia. Traditionally, hypoxemia is divided into anoxic anoxia (reduced hemoglobin saturation due to pulmonary disease), anemic anoxia (reduced hemoglobin, but normal saturation in carbon-monoxide poisoning), stagnant anoxia (normal hemoglobin saturation but reduced cerebral blood flow due to shock), and other less common forms. Moreover, at these low levels, blood pressure is unsustainable, and hypotension emerges; therefore, we often observe combined hypoxic/ischemic brain injury. We must distinguish purely hypoxic events from combined hypoxic/ischemic events—these very different stressors produce dramatically different pathogenesis and brain effects. Hypoxemia may cause syncope, but this requires very low arterial PO₂ (<30 mm Hg). When this is reached and maintained for minutes, neuronal injury can occur. As previously mentioned, these lesions typically appear in the more vulnerable globus pallidus. Hypoxemia eventually leads to hypotension and ischemia, which then trigger an apoptosis pathway resulting in irreversible damage. Patients without significant comorbidity tolerate brief, profound periods of hypoxemia without systemic acidosis or circulatory impairment. Isolated hypoxic brain injury damages deep gray matter such as the caudate, putamen, and thalamic regions; however, recovery is more complete than recovery following shock and hypoxemia.

Acute renal failure

Acute renal failure causes uremic-retention solutes. Guanidine compounds and glutamate rise significantly in patients with uremia. Glutamate

binds to N-methyl-D-aspartate and 2-amino-3 propanoic acid receptors. These neurotoxins cause clinically manifested uremic encephalopathy. Uremic encephalopathy produces inflammatory markers and disrupts microvascular barriers. Hypercoagulability in chronic and status renal disease induces a prothrombotic state and, potentially, cerebral infarction visible on MRI. Clinically, uremic encephalopathy results in marked drowsiness, dysarthria, occasionally interrupted by agitated delirium and, on examination, classical findings of asterixis, best described as a postural lapse rather than “flapping tremors” (postural loss does not resemble flapping, and sudden, arrhythmic postural loss is unlike any tremor). Asterixis is visible in the face, tongue (with protrusion), and all four extremities. Many nephrologists regard profound asterixis as an indication of dialysis. Asterixis poorly correlates to creatinine clearance and blood urea nitrogen (BUN) values.

Many patients with uremic encephalopathy develop PRES, particularly after a major hypertensive surge in patients with chronic kidney failure. The risk of a hypertensive emergency is substantial in chronic renal failure, but systolic blood pressures between 150 and 200 mmHg can damage the brain and kidney, resulting in a hard-to-break cycle of injury within these two connected systems. MRI shows hyperintensity in the white matter because of breakthrough edema. Originally considered limited to the posterior regions, it can extend throughout the brain. Others found basal-ganglia abnormalities in patients with extremely increased creatinine values, but whether this is a subset of the PRES clinical spectrum is uncertain [21].

Dialysis complications are uncommon, but in the past, rapid dialysis produced an acute syndrome known as “reverse urea hypothesis,” wherein significant urea gradients between blood and brain caused water influx to the brain. Other pathophysiological explanations were the “idiogenic osmol hypothesis,” where formation of idiogenic osmols led to increased venous osmolality and brain edema. Notably, post-dialysis recovery from uremic encephalopathy is typically slow, often lasting several days after renal-index normalization. The pathophysiological explanation is unclear as well as knowing when to start dialysis in a patient with acute, nonchronic renal disease. Without emerging sepsis or new structural lesions such as subdural hematoma or intracranial hematoma, dialysis can

be successful even when urea has not significantly increased from a prior baseline.

Acute liver failure

Ammonia remains the key driver of the pathogenesis of hepatic encephalopathy, and hepatic encephalopathy occurs when ammonia concentration exceeds 75 mmol/L. Animal models have shown that gamma-aminobutyric acid (GABA)-potentiating drugs cause hepatic encephalopathy. GABA is an alpha-aminobutyric acid [22]. Ornithine transcarbamylase (OTC) deficiency is an X-linked chromosomal defect and produces a urea-cycle disorder. Hyperammonemia may present in childhood or as late as the fifth decade. OTC is a mitochondrial enzyme converting ornithine and carbamyl phosphatase to citrulline, and its deficiency leads to hyperammonemia. Ammonia is cytotoxic, depleting intermediates of cell-energy metabolism, and most likely acts as an osmolyte causing astrocyte swelling. An opened blood-brain barrier accumulates glutamine, which, untreated, leads to cerebral edema. Computed tomography (CT) may show cerebral edema similar to fulminant hepatic failure, but few published cases exist. Plasma and urinary amino acid and urinary orotic-acid levels are necessary diagnostic tests as well as monitoring increased plasma citrulline or arginosuccinic acid. Emergency treatment of hyperammonemia includes continuous hemofiltration and intravenous administration of sodium phenylacetate and sodium benzoate, supplemented by arginine, citrulline, and carnitine. Dextromethorphan or other competitive inhibitors (e.g., kynurenic acid) are therapeutic options [23, 24].

The excitotoxicity of the neuronal glutamate receptors may best explain acute liver failure. The formation of glutamate, and possibly also aquaporin-4, produces nitric oxide and leads to cerebral edema and increased intracranial pressure. Astrocyte-detoxified ammonia through glutamine synthesis also causes osmotic stress. Cerebral edema requires immediate osmotic treatment in patients with acute hepatic failure. The success of organ transplantation directly correlates with aggressive control of brain edema and, thus, intracranial pressure [25].

Acute pancreatic disease

Sepsis, corticosteroids, or even high-dose, intravenous dopamine markedly increase serum glu-

cose with values reaching 800–1,000 mg/dl. Hyperglycemia in those ranges causes stupor and, eventually, unresponsive coma. Plasma osmolality generally correlates better with decreased consciousness levels. The nonketotic hyperosmolar state can display confusing focal signs. Some patients may have aphasia or a hemiparesis that improves after correction of glucose. In children, however, diabetic ketoacidosis and hyperglycemic-hyperosmolar state are life-threatening due to severe brain edema. Cerebral edema appears in children (and very rarely in adults) with diabetic ketoacidosis. Although the childhood incidence is <1%, its appearance boosts the mortality rate to 40%.

Patients with fatal cerebral edema display evidence of a disrupted blood-brain barrier [26]. As in liver failure, treating diabetic ketoacidosis may cause significant cerebral edema.

Acute thyroid failure

Acute thyroid failure affecting cerebral neuronal activity is less understood. The thyroid hormone influences cerebral neuronal activity, but T3 receptors localize mostly in the limbic system and amygdala rather than the brainstem, which contains the ascending reticular-activating system. Brain metabolism and nonspecific, reduced cerebral blood flow are found in hypo- and hyperthyroid states. The somewhat uncommon Hashimoto encephalopathy involves lymphocytic infiltration in the brain, meninges, cortex, basal ganglia, thalami, and hippocampus. MRI demonstrates periventricular white-matter lesions, which respond to corticosteroids; hence, the term steroid-responsive encephalopathy associated with autoimmune thyroiditis, also summarized as SREAT. Laboratory findings include a high-normal serum-thyrotropin level and increased thyroid-peroxidase antibodies [27–29].

Unexplained tachycardia or bradycardia associated, concomitant temperature changes in a comatose patient point to thyroid disease. Abnormally relaxed tendon reflexes (Chaney-Woltman sign best seen with the usually quick ankle reflex), pubic and axillary hair loss, and dry, puffy skin are important clues. Consciousness levels gradually become severely depressed. Treatment includes high-dose methylprednisolone, 500 mg daily for 5 days, and thyroxine, a 500- μ g loading dose followed by 1.8 μ g/kg daily.

These dramatically reverse somnolence within 24–36 h.

Acute cardiac failure

Cardiogenic shock with extremely low blood pressure requiring extracorporeal membrane oxygenation (ECMO) cause diffuse hypoxic-ischemic brain injury. Predominantly involving the putamen, cerebellum, and hippocampi, it leads to persistent chroma following correction of cardiac failure. Cardiac and circulatory arrest are defining factors and will, despite successful cardiopulmonary resuscitation (CPR), lead to global injury in most patients. Coma after post-CPR resumption of circulation correlates with severe anoxic-ischemic injury to the thalamus, striatum, and globus pallidus. Additionally, it can injure the cortical mantle and its multiple layers (also known as laminar necrosis). Several factors determine poor outcomes. These include early anoxic brain swelling; diffuse cortical restriction on diffusion-weighted MRI; absent cortical somatosensory evoked potentials (SSEP); rising serum neuron-specific enolase values; unreactive, burst-suppression EEG patterns or marked suppression to $<20 \mu\text{V}$; and burst-suppression EEG time locked with myoclonus status. Improvement usually happens during the same hospitalization. Confounders significantly complicate prognostication of comatose patients after successful CPR. Drug washout in a severely afflicted patient with post-CPR-associated kidney and liver injury remains difficult to assess. Conspicuously lacking in many studies on outcome are details about the neurologic examination, use of sedation, neuroimaging, and post-resuscitation hemodynamic and organ function, often crucial in deciphering prognoses. Circulatory arrest is the most severe, perhaps most devastating, injury to the brain. Lack of immediate availability of MRI and SSEP hampers prognostication. Many patients have myoclonal status and electrographic seizures, but a recent study has shown that aggressive management does not improve outcome [30, 31].

Multiple organ failure

Sepsis is the most recognized cause of multiple organ failure, and damage to the brain is likely caused by simultaneous dysfunction of two or more organs. We do not fully understand how sepsis damages the brain or whether peripheral cytokines alter the blood–brain barrier. Neuropathological brain studies of patients who died from sepsis show accelerated, microglial activation-associated

neuronal apoptosis in several areas but significantly in the locus coeruleus, part of the ascending reticular-activating system and crucial for wakefulness. Sepsis-associated inflammatory processes do not always translate to brain injury. Most likely, sepsis-associated multi-organ failure causes impaired cerebrovascular autoregulation and decreased cerebral blood flow, but the culprit might also be hypotension and/or anoxic injury [32]. A separate, unresolved issue is thermal strain to the brain when fevers reach 39°C ; its correlation with apathy, delirium, or stupor; and its potential resolution when fever breaks. Structural injury post sepsis is present but unclearly related. An MRI study of 72 patients with profound encephalopathy or coma and septic shock detected only nonspecific white-matter changes and prior, small territorial strokes but was normal in half the cases [33].

Metabolic toxic intermediates and cofactor deficiencies

Clinicians may consider this diagnostic category only in certain circumstances: (1) an unusual history, (2) an unusual clinical course, or (3) an unusual MRI. While there are numerous genetic metabolic defects, few appear for the first time in adulthood. This rare category of disorders yields four causes of encephalopathy: acute porphyria, pellagra, thiamine deficiency, and MELAS (mitochondrial encephalomyopathy, lactic acidosis, and stroke-like episodes). We find clues to these disorders by measuring serum lactate (for lactic acidosis), serum ammonia (hyperammonemia), and severe malnutrition (recent IV-glucose administration to uncover a severe B1 deficiency). Clinicians should consider these disorders under the appropriate circumstances (Fig. 2).

Acute intermittent porphyria (AIP) is an autosomal-dominant disorder caused by a partial deficiency of uroporphobilinogen (PBG) deaminase, the third enzyme in the heme synthesis pathway. The condition remains quiescent until precipitated by stress, menstruation, fasting, pregnancy, surgery, or medications. During an acute attack, heme precursors, such as aminolevulinic acid (ALA) and PBG, discharge in large quantities in urine. Suggested mechanisms include a direct neurotoxic effect of ALA, which is structurally similar to GABA and interacts with GABA receptor. ALA also crosses the blood–brain barrier easily, accumulates in neuronal tissue, and alters tryptophan metabolism. AIP-associated MRI abnormalities resemble PRES with involvement of the

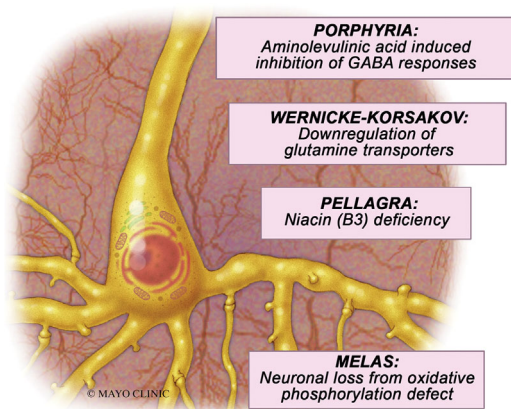


Fig. 2 Illustration of some toxic intermediates causing neuronal damage.

occipital or frontoparietal cortices (anterior lesions also appear in 90%). Although lesions are readily reversible, impaired microcirculation and tissue ischemia with protracted or recurrent episodes may cause cerebral infarction. Other abnormalities include central pontine myelinolysis due to AIP-associated hyponatremia [34]. No comprehensive data exist on imaging accompaniments of porphyric encephalopathy, although most episodes likely involve transient or permanent MRI changes. Seizure causes in porphyric encephalopathy are unclear. Suggested mechanisms include a direct neurotoxic effect of ALA, structurally similar to GABA and interacting with GABA receptors. ALA also crosses the blood–brain barrier easily, accumulates in neuronal tissue, and alters tryptophan metabolism. Treating patients with severe, acute, porphyria-associated encephalopathy requires immediate administration of hydroxylated heme (hemin), administered intravenously, 2–5 mg/kg daily, for 14 days or until improvement is noted.

Adult hyperammonemia often correlates with laboratory evidence of acute or chronic liver failure. As an isolated laboratory abnormality it is rare. Defects in the urea cycle are often present. Its occurrence in the immediate postoperative course of lung transplantation has been well recognized and may be a cause of unexplained stupor or seizures (34A). Some have found disseminated *Ureaplasma* species as cause of this syndrome (with improvement after tetracycline), others have attributed to a urea cycle defect unveiled by the

stress of the operation or total parenteral nutrition but it has remained an unexpected mystery in many cases (34B).

Pellagra should be considered in patients with alcohol dependence, particularly with unexplained persistent encephalopathy, neurological symptoms, gastrointestinal distress, or diffuse epidermal scaling with erythema of exposed skin resembling seborrheic dermatitis. Homeless populations are vulnerable, since poor diet and high incidence of alcohol-use disorders increase the risk of pellagra [35]. While awaiting serum-niacin results, oral nicotinamide, 100 mg three times a day, should be initiated.

Depleted thiamine stores occur within 3 weeks in patients with a thiamine-free diet [36]. Glucose infusion activates glycolysis, which consumes vitamin B1 and depletes thiamine stores. Thiamine deficiency triggers neurological abnormalities, but specific areas of neuronal loss are more complex, possibly mediated through histamine or down regulation of glutamate transporters [37].

Other neurological disorders associated with deficiencies of water-soluble B vitamins involve the spinal cord rather than the brain. MRI shows bilateral hyperdensity in mamillary bodies, medial thalami, periductal gray as well as in the pons, medulla oblongata, and frontoparietal cortex [38]. Management of Wernicke–Korsakoff syndrome requires immediate administration of thiamine, 100 mg intravenously, but also magnesium and additional multivitamin supplements. Improved consciousness can occur in some patients but is protracted or minimal when MRI abnormalities are profound. Persistent, long-standing cognitive abnormalities and ambulation difficulties due to severe ataxia are expected [39].

MELAS is an acronym used to describe a distinctive clinical syndrome that eventually presents with mitochondrial myopathy, encephalopathy, lactic acidosis, and stroke [40]. Certain, common tell-tale signs are seizures, stroke-like episodes, short stature, and mental retardation, but muscle weakness, headaches, and easy fatigability are also nonspecific presenting symptoms. Cortical involvement is prominent on MRI and sparing of the deep white matter, explained by a higher metabolic turnover in the cortex, which also makes it more vulnerable to injury. The neuropathology of MELAS is extensive neuronal loss, neuronal eosinophilia,

Table 2. Initial approach to encephalopathy

1	Document detailed neurologic findings and grade level of consciousness.
2	Obtain new basic metabolic panel and trend for substantial changes.
3	Obtain arterial blood gas.
4	Obtain serum ammonia and lactate.
5	Obtain computed tomography scan or magnetic resonance imaging scan of the brain (vascular studies when appropriate).
6	Consider 24-h EEG monitoring.
7	Obtain CSF for inflammatory or infectious markers (obligatory in any immunocompromised patient).
8	Replace electrolytes and correct vitamin deficiencies (when indicated).
9	Discontinue or replace antimicrobials (if toxicity is probable).
10	Consider genetic consultation and additional focused tests.

vacuolization, and astrogliosis. The abnormalities also extend beyond the typical vascular territories. Intriguingly, basal ganglia are involved with globus pallidus and caudate nucleus abnormalities in approximately half the patients with calcification on CT [41–44]. MELAS involves a deficiency in the mitochondrial respiratory chain complex. Diagnosis is based on mitochondrial DNA analyses of peripheral blood showing A→G point mutation at position 3243. Muscle biopsy could demonstrate so-called “ragged red muscle fibers,” a signature of all mitochondrial diseases. MELAS patients have increased serum lactic acid, but only an increased CSF lactate value correlates with outcome [41]. Coenzyme Q10 may restore the defect in oxidative phosphorylation due to electron transfer from complexes 1 and 2 to 3. MELAS patients usually receive coenzyme Q10 (2 mg/kg/day) for 6 months [40].

Conclusions

Encephalopathies confront physicians already in a quandary from (1) the sheer number of causes and (2) definitional uncertainties. Fortunately, physicians in most clinical practices need only consider a few causes. Usually, a focused but comprehensive evaluation identifies the cause. Table 2 offers a reasonable initial approach with necessary ancillary tests and serious considerations of others.

The term *metabolic encephalopathy* may have been mistakenly carried over to the age of modern neuroimaging. The recognition of structural lesion damage in metabolic encephalopathy was an important breakthrough. Metabolic encephalopathies are an unfinished medical narrative. Currently, clinicians (and, particularly, consulting neurologists) use words with several meanings and ambiguity about which meaning is intended. Two experienced specialists may examine the same patient and arrive at different conclusions and treatment recommendations. For example, does this level of BUN explain the clinical picture, or should we consider a confounder? Does this level warrant dialysis? We cannot declare a moratorium on the term metabolic encephalopathy, but we also cannot consider it uncritically because it is embedded in hospital practices. Nevertheless, we must confront the ambiguities to improve the definitions for future clinical studies, pathophysiologic understanding, and, ultimately, our practices. All of these require comprehensively studied prospective cohorts of patients with an initial tentative diagnosis of encephalopathy.

For now, our clinical approach must determine which organ dysfunction is clinically and physiologically relevant. In the absence of MRI evidence of structural lesions, normal cerebrospinal fluid formula, and seizures on prolonged EEG monitoring, we should assume that patients can recover once the organ failure resolves. Lack of recovery requires another look and another series of focused tests while considering “rarities,” many of which are afterthoughts.

Conflict of interest

No conflict of interest was declared.

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